

Appendix

Prototypes

The following design prototypes (Figure 6 – 29) are part of the iterative development process of our study. Each prototype represents a distinct approach to simplifying and translating the multi-dimensional aspects of the data into easily interpretable values.

Single value 1

80%
of the maximum reachable points

4

FIGURE 6. Maximum Reachable Points (Single Value 1):

This visualization prominently displays 80% of the maximum achievable points using a straightforward and bold design. It offers a clear and concise depiction of progress, which is advantageous for quickly assessing performance. However, it does not provide additional context or metrics. By presenting data in an easily understandable format, it aligns with project goals focused on inclusivity and accessibility. While it simplifies numerical complexity and uses an intuitive metric familiar to most audiences, it does not offer the detailed insights required for more in-depth analysis.

Single value 2

80
points out of 100

5

FIGURE 7. Absolute Points (Single Value 2): This visualization shows a score of 80 points out of a total of 100, offering a straightforward and easy-to-understand representation. It clearly indicates both the achieved score and the total possible points, making it a transparent choice for users. However, it does not facilitate comparison or benchmarking against other scores. By clearly communicating the raw score and total, this design aligns with project goals by providing clarity and simplicity for users who favor direct information. While it reduces ambiguity and builds trust by displaying both the score and total, it lacks context for users less familiar with quantitative assessments, such as whether the score is "good" or "bad."

Single value 3

20 points
going from -60 to 40

6

FIGURE 8. Absolute Points on Explicit Scale (Single Value 3):

This visualization highlights a score of 20 points within a specified range from -60 to 40, providing insight into the score's position on a defined scale. While it aids in understanding score relativity, the explicit range might need further explanation for some users, especially due to its skewed nature. By emphasizing scale, it supports users in contextualizing their performance relative to the possible range, aligning with project goals. However, it may present challenges for those less familiar with such scales, affecting its inclusivity.

Single value 4



7

FIGURE 9. Quality Spectrum (Single Value 4): This figure uses a red-to-green colored bar to depict quality levels, with a marker indicating a position within the "good" range. It leverages intuitive color-coding to communicate quality, aligning with familiar performance indicators. While this method enhances accessibility for users with minimal data literacy, it oversimplifies more complex evaluations, proving to be unsuitable for stakeholders.

Single value 6



9

FIGURE 11. Quantified Range (Single Value 6): Combining a gradient bar with a numerical range from -60 to 40 and a marker at 17, this design merges qualitative and quantitative insights, offering more precise interpretation within a range. While it effectively caters to diverse user preferences by providing both visual and numerical data, the complexity might overwhelm those unfamiliar with such presentations, potentially necessitating additional guidance to enhance inclusivity.

Single value 5



8

FIGURE 10. Risk Spectrum (Single Value 5): Depicting risk levels on a gradient from "high risk" to "low risk," this visualization provides a visually intuitive assessment, aided by a color gradient for rapid interpretation. While it aligns with the goal of evaluating data use impact, it lacks precise quantification needed for detailed risk management. The simplicity of the gradient enhances accessibility but might not suffice for users needing precise, actionable guidance.

Single value 7



10

FIGURE 12. Step-wise Gradient (Single Value 7): This visualization features a gradient bar from light to dark purple, marking a score of 83 out of 100 with discrete steps. It is visually appealing and straightforward, clearly indicating the proportion of the total score achieved. While it excels in clarity with its structured approach, it might lack additional context or benchmarks. This design supports project goals by providing segmented clarity, appealing to users who prefer distinct steps. However, its simplicity may not satisfy advanced users seeking detailed insights.

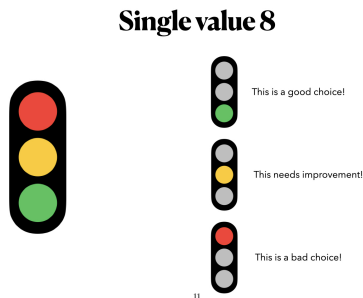


FIGURE 13. Traffic Light Indicator (Single Value 8): Employing a traffic light metaphor with green, yellow, and red states, this visualization provides a universally recognizable and intuitive evaluation of success or concern. While it simplifies decision-making through categorical grouping, it may lack nuance. By ensuring instant comprehension and minimal numerical demands, it aligns with inclusivity goals, though it may not satisfy users seeking detailed analysis.

Single value 10

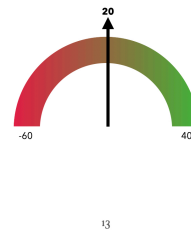


FIGURE 15. Half Arc with Range (Single Value 10): This visualization combines a semi-circular gauge with a numerical range of -60 to 40, marked at 20. It merges numerical detail with a visual indicator, intuitive for assessing performance within a defined range. While it balances visual appeal with information, interpretation might need further explanation for less data-literate users, especially regarding the range's significance.

Single value 9

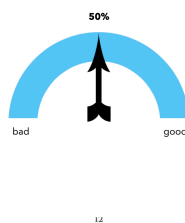


FIGURE 14. Half Arc Indicator (Single Value 9): Featuring a semi-circular gauge with an arrow at 50%, labeled from "bad" to "good," this design is visually dynamic and engaging. It effectively represents a balance or midpoint measure, offering an intuitive sense of relative performance. While aesthetically appealing for a wide audience, the exact figures are slightly obscured in this representation.

Detail 1

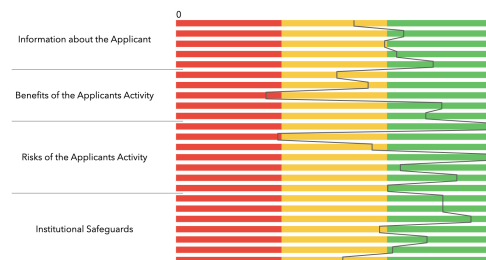


FIGURE 16. Gradient Horizontal Bars with Line Overlay (Detail 1): This visualization features horizontal bars segmented into red, yellow, and green to represent negative, neutral, and positive results, with a line graph overlay illustrating trends. It highlights variations and categories, offering actionable insights. However, the line overlay may complicate interpretation for some users. The design aligns with project goals by exploring relationships among dimensions and requires careful attention to ensure clarity for all users.

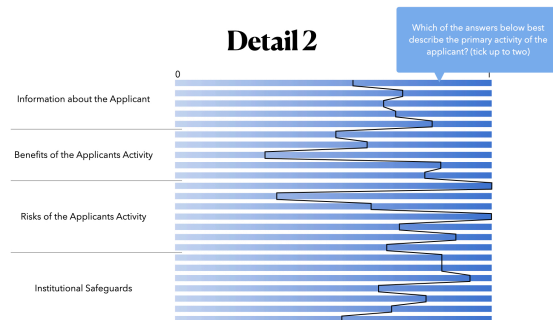


FIGURE 17. Gradient Bars with Prompt Overlay (Detail 2): Similar to Figure 16, this chart uses a blue gradient background and includes a textual prompt for guidance. It enhances interpretability with context but may rely too heavily on the gradient, potentially overshadowing detailed insights. By providing explanatory prompts, it bridges understanding gaps and aligns with inclusive goals, though supplementary explanations might be necessary for complete clarity.

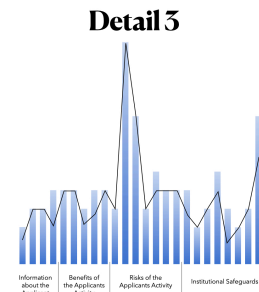


FIGURE 19. Vertical Bar Chart with Line Overlay (Detail 3 - Type 2): Featuring vertical bars for each category with a line graph overlay, this design simplifies comparison and adds trend-based analysis. While effective for longitudinal assessments, it may become visually dense. The familiar bar structure ensures accessibility, though simplified annotations might help prevent overwhelming users.

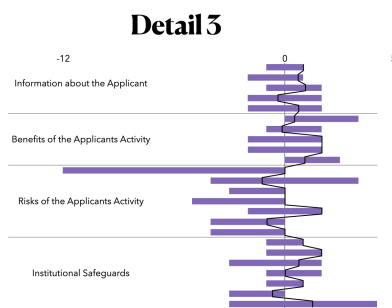


FIGURE 18. Horizontal Bars with Diverging Axes (Detail 3 - Type 1): This chart presents horizontal bars diverging from a central axis, with an overlaying line graph to show trends. It clearly distinguishes between negative and positive metrics but may increase cognitive load for users unfamiliar with this format. The design supports nuanced exploration of imbalances, aligning with project goals, but additional labeling or legends might be needed to enhance understanding.

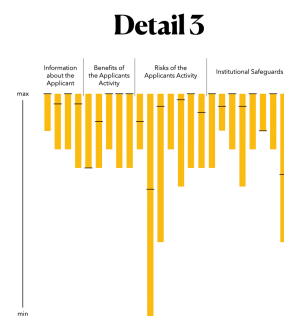


FIGURE 20. Inverted Bars with Max/Min Indicators (Detail 3 - Type 3): This visualization uses vertical bars extending downward from a central axis, marking maximum and minimum values. It emphasizes the range of values but may confuse users unfamiliar with inverted scales. By highlighting extremes, it supports identifying critical thresholds and risks, aligning with project goals, though annotations may be necessary to guide interpretation.

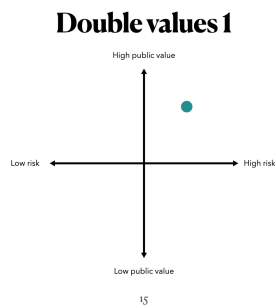


FIGURE 21. Basic 2D Grid (Double Values 1): This simple Cartesian plane is divided into quadrants with axes representing "Public Value" and "Risk," plotting a single data point to indicate the respondent's result. The straightforward design is easy to interpret, offering clear categorization (e.g., high public value and high risk). While lacking in contextual elements like colors or ranges, its minimalism ensures clarity and accessibility for those with lower data literacy, aligning with inclusivity goals by avoiding complex visuals. However, it may not provide deeper insights into trends or distributions.

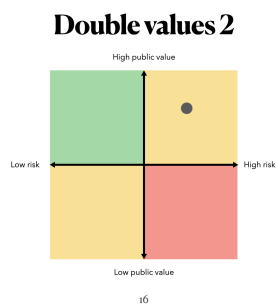


FIGURE 22. Quadrants with Color Coding (Double Values 2): Building on the basic 2D grid, this design incorporates color coding for each quadrant, such as green for "Low Risk/High Public Value" and red for "High Risk/Low Public Value." This enhances interpretability and offers immediate visual feedback, helping categorize results into favorable or unfavorable zones. While intuitive for most users, the design may not be accessible for colorblind users without alternative indicators, and it might oversimplify nuanced relationships between public value and risk. The design illustrated by this prototype was ultimately chosen for the final version of the tool in combination with complementary visual components to balance out the discussed shortcomings.

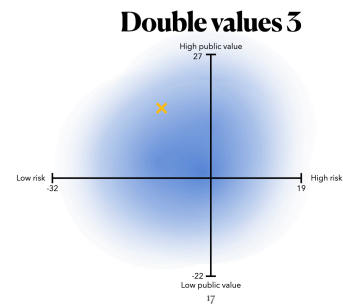


FIGURE 23. Heatmap Overlay (Double Values 3): This visualization merges a Cartesian plane with a heatmap, using a background color gradient to indicate data density, with a yellow "X" representing the respondent's position. It adds richness by highlighting trends or clusters, providing context on how a result compares to others. While visually engaging, it may overwhelm non-technical audiences, requiring supplementary guidance for full inclusivity. For data-savvy users, however, the additional context enhances interpretability and engagement.

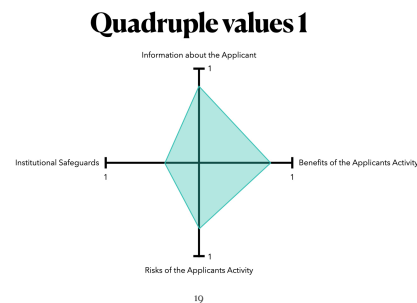


FIGURE 24. Radial Diagram (Quadruple Values 1): This radial chart connects values across four axes to form a polygon, intuitively visualizing balance or disparities between dimensions. It highlights strengths and weaknesses effectively, although it may be challenging for those unfamiliar with radial charts. By conveying a multidimensional perspective, it aligns with project goals of holistic and inclusive data interpretation. While engaging for varied cognitive preferences, unfamiliar users may benefit from annotations or a legend to aid understanding.

Quadruple values 2

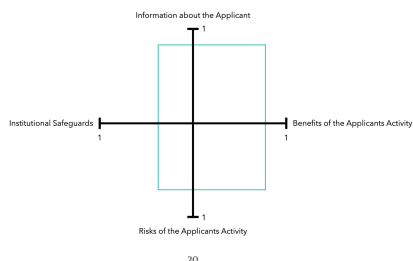


FIGURE 25. Rectangular Overlay (Quadruple Values 2):

This design features four axes radiating from a central point to form a rectangle, indicating clear boundaries for maximum and minimum values. Simpler than radial charts, it reduces cognitive load but may lack proportional representation. It effectively communicates data limits and comparative utility, aligning with simplicity and usability goals. The minimalistic approach enhances accessibility, appealing to diverse visual literacy levels without overwhelming complexity.

Quadruple values 4

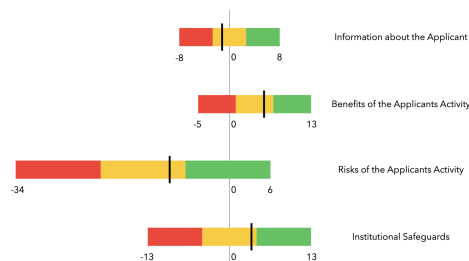


FIGURE 27. Horizontal Bar Chart with Color Coding (Quadruple Values 4):

This chart uses four horizontal bars to represent each dimension, with sections color-coded to indicate positive, neutral, and negative values. It offers a straightforward comparison between dimensions, enhancing interpretability through color coding. While simplifying relationships, it requires colorblind-friendly palettes for accessibility. This design reduces complexity, supports clear categorization, and aids quick interpretation, aligning with project goals and inclusive data experience.

Quadruple values 3

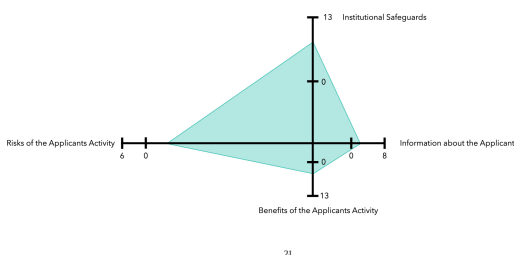


FIGURE 26. Irregular Polygon with Scaled Axes (Quadruple Values 3):

A radar chart with independently scaled axes forms an irregular polygon, emphasizing relative performance in each category. While offering flexibility for different ranges, it may confuse users with its separate axis scales and potential distortion in proportional comparisons. By accommodating varied data ranges, it supports detailed analysis and aligns with project goals for analytical depth. However, additional explanations may be needed to ensure inclusivity and prevent misinterpretation.

Quadruple values 5

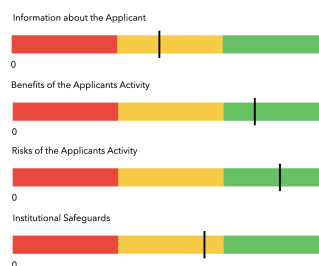


FIGURE 28. Standardized Horizontal Bar Chart (Quadruple Values 5):

Similar to the previous chart, this version normalizes scales (0–1) for consistency across dimensions, simplifying comparisons. The uniform scale facilitates direct comparison but may obscure the magnitude of differences. By maintaining balance and standardization, it aligns with the project's goals for equal data representation and offers an engaging, simple visual format for varied expertise levels.

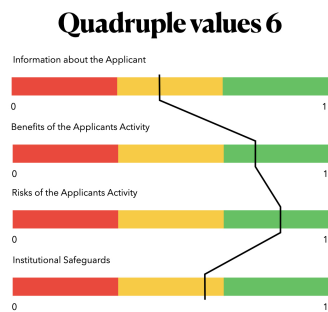


FIGURE 29. Linked Bar Chart (Quadruple Values 6): This chart features horizontal bars for each dimension, connected by a line to illustrate relationships or flow between values. It highlights patterns and correlations, adding relational insight. While this approach facilitates advanced analysis and aligns with project goals of capturing complex interdependencies, it requires clear annotations to avoid misinterpretation and ensure accessibility, especially for users less familiar with relational visuals.

Usability Study Supplements

Data Use Practices of Diagnosys AI

Scenario: Evaluate a proposed project where Diagnosys AI, a private company, uses patient data from the public Capital Region Hospital Network.

Activities: Diagnosys AI will analyze pseudonymized X-ray/CT scans to conduct research improving their AI, aiming to sell a product/service (diagnostic software).

Funding & Reporting: Diagnosys AI, a medium-sized entity, is funded by income from its own activities (revenue) and a grant from a public body (financing by governmental organisations). They provide an annual public report on project activities (not financials) for the grant. General public access to information relies on GDPR (no legal options to compel disclosure beyond civil or criminal proceedings). They have a procedure in place to respond to requests for information from members of the public (data subjects) seeking access to, or information about the processing of, their personal data.

Motivation & Data: Uses historical, pseudonymized images. Stated motivations are research (industry-focused) and business and commerce.

Impact: The company argues for benefits covering current and future relevance and that benefits are likely to increase in the future, accruing mainly to Diagnosys AI (commercial) and healthcare provider customers, with potential gains for patients. They use cloud providers with some measures to limit environmental impact.

Benefit Distribution: Focused on high-income countries; no mechanism or intention to benefit low- and middle-income countries (LMICs). Past critiques noted slightly lower AI accuracy for some demographics; they have some, but not a strong or consistent, track record of proactively strengthening benefits for protected groups.

Risks: Foreseeable risks include informational risk to data subjects (the supposedly anonymous data of the patients could potentially be linked back to them), health risk to future patients (from potential AI misdiagnosis), and social risk (AI bias potentially worsening health disparities). Overall likelihood is moderate foreseeable risk for all categories. The data use foreseeably entails elevated risk for protected groups, meaning individuals identified by characteristics like race or gender are more likely to be negatively impacted, potentially due to biases in the data or AI model. There is no information about the specific risks of this project proactively shared with the general public. No specific steps are described to prevent misuse of the

algorithms or insights.

Safeguards: Diagnosys AI uses ad-hoc risk assessments internally; findings usually result in information for technical teams. The process involves technical staff and legal experts. There is no dedicated process described for monitoring broader harms post-deployment. Stopping the software use across the network cannot be ended immediately. Patients complain via the hospital; Diagnosys AI has no direct public complaint procedure. Complaints escalated to Diagnosys AI are acknowledged.

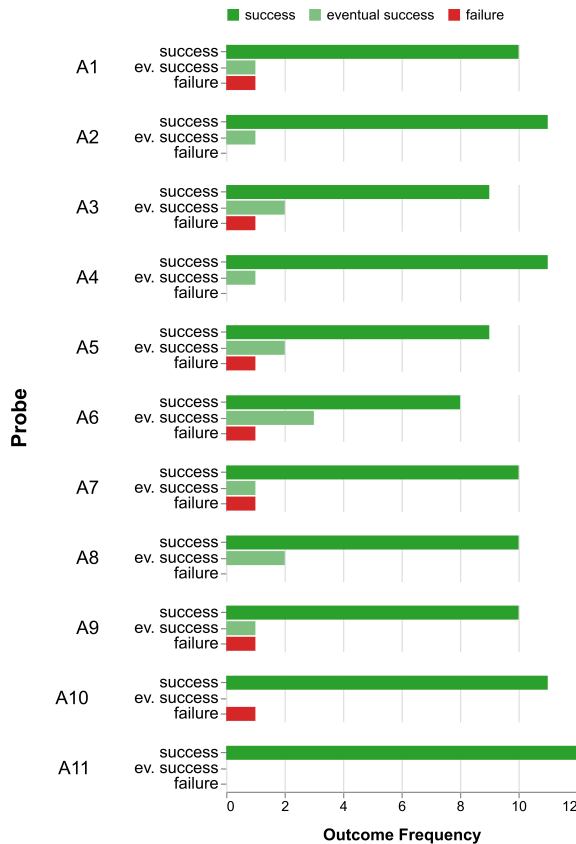


FIGURE 30. Participant comprehension outcomes ($N = 12$) for the 11 structured probes administered during the results interpretation task. Each probe ID (A1-A11) corresponds to the questions detailed in Table 2. The results show high overall comprehension, with a majority of participants achieving immediate success on most tasks. Notably, all 12 participants successfully synthesized the information to provide an overall summary (A11), indicating the tool's effectiveness in communicating its final assessment.

TABLE 2. Probes to guide the results interpretation session.

#	Moderator Probe	Purpose
A1	<i>Initial Impression</i> First look – what is this page telling you overall?	Assess initial grasp and focus.
A2	<i>Visualization Comprehension</i> Where does the result show up on the chart?	Test basic chart orientation.
A3	What do 'Risk' and 'Benefit' on the axes mean here?	Gauge comprehension of the core risk-benefit framework.
A4	What does the dot's position in this quadrant suggest?	Test understanding of the visual classification system.
A5	What about these specific numbers for Risk and Benefit?	Check if users link scores to the visual display.
A6	And this 'Type [X]' classification – what does that imply?	Evaluate comprehension of the data solidarity typology.
A7	<i>Input-Output Link</i> This text mentions '[X] answers impact risk...' – what does that mean?	Assess if users understand how their inputs generated the output.
A8	<i>Recommendation Comprehension</i> Let's look at a recommendation... What is this first one suggesting?	Test the clarity and directness of the textual advice.
A9	<i>Recommendation Perceived Value</i> Is that clear? Does it seem relevant to the scenario? Useful?	Gauge perceived actionability and relevance of advice.
A10	<i>Utility Feature Findability</i> How would you save or share these results?	Evaluate findability of key utility features.
A11	<i>Overall Summary</i> In your own words, what's the main outcome for the project according to this?	Assess ability to synthesize information into a conclusion.

TABLE 3. Subjective usability rating questions.

#	Questionnaire Item	Purpose
B1	Overall, how difficult or easy was it to use the PLUTO tool? (1=Very Difficult, 5=Very Easy)	Capture perception of effort and interaction fluency.
B2	How clear or unclear were the questionnaire questions? (1=Very Unclear, 5=Very Clear)	Measure comprehensibility of survey content.
B3	How easy or difficult was it to understand the results visualization? (1=Very Difficult, 5=Very Easy)	Assess interpretability of the main visual output.
B4	How clear or unclear were the textual recommendations? (1=Very Unclear, 5=Very Clear)	Evaluate legibility and directness of actionable advice.
B5	How useful or not useful do you think those recommendations would be? (1=Not at all Useful, 5=Very Useful)	Gauge perceived value and applicability of the tool's guidance.
B6	How confident are you in understanding these results? (1=Not at all Confident, 5=Very Confident)	Measure self-assessed certainty in interpretation.

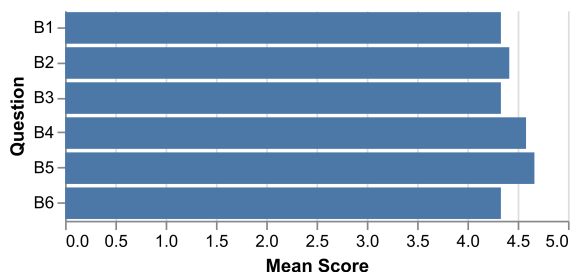
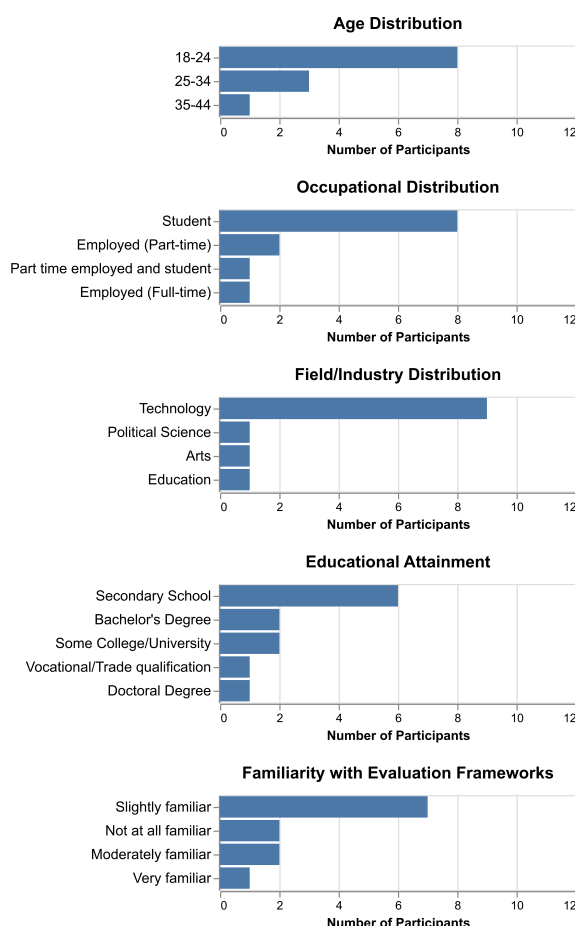
**FIGURE 31.** Mean user ratings ($N = 12$) on a 5-point Likert scale for key aspects of the PLUTO tool. Each question ID (B1-B6) is detailed in Table 3. The consistently high scores across all dimensions provide quantitative support for the tool's usability and effectiveness.**FIGURE 32.** Demographic distribution of the $N = 12$ study participants. The data shows participant details across age, occupation, field of work, educational attainment, and prior familiarity with evaluation frameworks. Crucially, the results confirm the cohort was composed of non-specialists, with a majority reporting little to no prior experience in assessing projects for societal impact.

TABLE 4. System Usability Scale (SUS) items, rated on a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree).

#	Statement	Purpose
Q1	I think that I would like to use this system frequently.	To calculate a standard usability score.
Q2	I found the system unnecessarily complex.	
Q3	I thought the system was easy to use.	
Q4	I think that I would need the support of a technical person to use this system.	
Q5	I found the various functions in this system were well integrated.	
Q6	I thought there was too much inconsistency in this system.	
Q7	I would imagine that most people would learn to use this system very quickly.	
Q8	I found the system very cumbersome to use.	
Q9	I felt very confident using the system.	
Q10	I needed to learn a lot of things before I could get going with this system.	

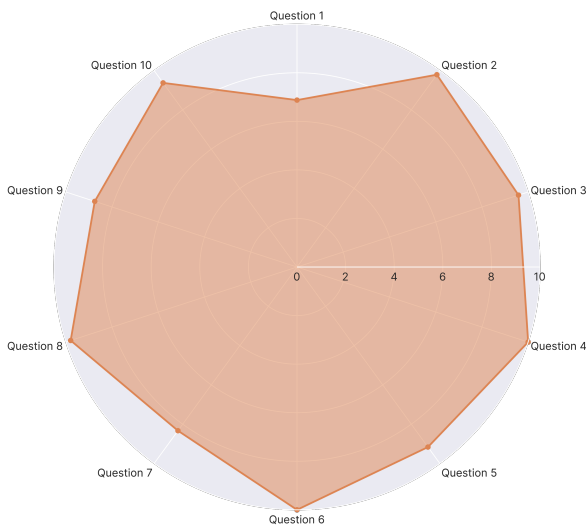


FIGURE 33. Per-question mean scores for the SUS questionnaire (see Table 4 for question text). The strong positive ratings for items related to ease of use (Q3), confidence (Q9), and learnability (Q7) align with the high overall score. The lower rating for Q1 ("I think that I would like to use this system frequently") was consistently explained by participants as a reflection of the task's specific, infrequent nature, not as a critique of the tool itself.



FIGURE 34. Distribution of System Usability Scale (SUS) scores from the 12 participants. The mean score of 91.67 places the tool in the "Best imaginable" range for perceived usability, with a tight distribution indicating consistently positive experiences.

TABLE 5. Probes for the concluding semi-structured interview.

#	Moderator Probe	Purpose
C1	What was your overall impression of the PLUTO tool?	Elicit a summary judgment and top-of-mind thoughts.
C2	What was particularly easy? What was most difficult?	Identify specific strengths and weaknesses in the UI and flow.
C3	Did the questionnaire options fit the scenario well?	Assess perceived completeness and applicability of content.
C4	Based on using it, what is the main purpose of PLUTO?	Check if the tool's core value proposition is communicated.
C5	Who do you imagine using this tool? Could a non-expert use it?	Gauge perceived inclusivity and accessibility for the target audience.
C6	Do you have any final comments or suggestions for improving PLUTO?	Collect user-driven ideas for future design iterations.

TABLE 6. Selected thematic quotes from the usability study, illustrating the core user experiences.

Theme	Quote	Relevance
Agency & Empowerment	<i>"My takeaway as a patient is that I would not give them my data because it's discouraged... Personally, the outcome for me would be to not give them my data."</i>	Shows the user making a definitive, personal decision based on the tool's assessment, demonstrating direct empowerment.
	<i>"If this were my company, I would have to think heavily about better carving out and communicating the future benefits of this company."</i>	The user adopts a decision-maker's perspective, considering concrete actions for project improvement based on the tool's results.
	<i>"If I was evaluating it from a government perspective, I would probably look at these recommendations and maybe try to implement some thresholds..."</i>	Illustrates empowerment at a policy level, where the user envisions using the tool's output to inform regulation.
Explainability & Trust	<i>"The tool affirmed what I initially had in mind... So I feel pretty confident that the tool aligned with what I initially had in mind."</i>	Shows trust being built when the tool's logic validates the user's own intuition about the scenario.
	<i>"That reminds me of the question where I was like, 'Oh, research for industry or for science..?'... if I had clicked science, that probably would have been counted as a bigger benefit."</i>	Demonstrates the user understood the causal link between their specific input and the final score, a core aspect of explainability.
	<i>"I would expect the company that wants my data to give me this information and proof... I would expect a filled out Pluto survey from them."</i>	The user reframes the tool as a mechanism for corporate transparency, showing trust in its framework to hold others accountable.
	<i>"I would kind of like to know how the recommendations are generated."</i>	User curiosity about the tool's logic indicates trust in the output.
Learning & Reflection	<i>"I was somewhat surprised... I still perceived the value to be quite high... But I guess the way the tool works, it classified it to be lower because of the harm... When I think about it now, it makes sense to me."</i>	Captures a moment of reflection where the user re-evaluates an initial bias after the tool highlights a previously overlooked factor.
	<i>"They haven't really, really thought about how they use the data in depth... It seems like they are thinking a lot about their technology; that's what I understand from the results."</i>	Illustrates a key takeaway where the user infers the project's priorities (technology over societal impact) from the assessment results.
	<i>"Although the risk... isn't that high and the benefit... isn't that low, the combination... makes it discouraging. It has too many risks for the quite low benefit."</i>	The user articulates the core risk-benefit trade-off, demonstrating a grasp of the tool's central evaluative concept.
Usability & Interpretation	<i>"I would have expected the edit button to either take me directly to the question... it made me feel like, 'Oh, well, that's just kind of pointless.'"</i>	Provides specific, actionable feedback on a UI feature whose behavior violated user expectations, causing friction.
Visualization Comprehension	<i>"The best would be having high benefit and low risk, which would be somewhere in this top-left corner. And the worst would be if we have a use case where there is high risk and low benefit."</i>	The user correctly interprets the 2D risk-benefit matrix, including the ideal and worst-case quadrants.
	<i>"It could initially be a little bit confusing that low risk is good and high risk is bad... But the coloring makes it very intuitive for me."</i>	Identifies a potential confusion with axis orientation but confirms that a specific design choice (color-coding) effectively resolved it.